

Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas

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Tarea 4

Definition 1. Let a C^2 function f be defined on some domain $X \subset \mathbb{R}^n$, and let $x \in X$. Denote the second partial derivative of f as $f_{ij} := \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$.

For $k = 1, \dots, n$, let $\bar{H}_k[f(x)]$ be the $(k+1) \times (k+1)$ matrix given by

$$\bar{H}_k[f(x)] := \begin{bmatrix} 0 & f_1(x) & \cdots & f_k(x) \\ f_1(x) & f_{11}(x) & \cdots & f_{1k}(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_k(x) & f_{k1}(x) & \cdots & f_{kk}(x) \end{bmatrix}.$$

The above matrix is the leading principal minor of the **bordered Hessian** of f at x of order k . The following theorem gives the second-derivative characterization of a *quasiconcave* function.

Theorem 1. Let X be an open and convex set in $\mathbb{R}^n$, and let $f : X \rightarrow \mathbb{R}$ be a C^2 function. Then

(a) If f is quasiconcave on X , then we have $(-1)^k \det(\bar{H}_k[f(x)]) \geq 0$ for $k = 1, \dots, n$;

(b) If $(-1)^k \det(\bar{H}_k[f(x)]) > 0$ for $k = 1, \dots, n$, then f is quasiconcave on X .

Exercises

Ex. 1.12 Suppose $u(x_1, x_2)$ and $v(x_1, x_2)$ are utility functions.

(a) Prove that if $u(x_1, x_2)$ and $v(x_1, x_2)$ are both homogeneous of degree r , then $s(x_1, x_2) \equiv u(x_1, x_2) + v(x_1, x_2)$ is homogeneous of degree r .

(b) Prove that if $u(x_1, x_2)$ and $v(x_1, x_2)$ are quasiconcave, then $m(x_1, x_2) \equiv \min\{u(x_1, x_2), v(x_1, x_2)\}$ is also quasiconcave.

Ex. 3.C.6 Suppose that in a two-commodity world, the consumer's utility function takes the form

$$u(x) = [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho}.$$

This utility function is known as the *constant elasticity of substitution* (or *CES*) utility function.

- (a) Show that when $\rho = 1$, indifference curves become linear.
- (b) Show that as $\rho \rightarrow 0$, this utility function comes to represent the same preferences as the (generalized) Cobb–Douglas utility function $u(x) = x_1^{\alpha_1} x_2^{\alpha_2}$.
- (c) Show that as $\rho \rightarrow -\infty$, indifference curves become “right angles”; that is, this utility function has in the limit the indifference map of the Leontief utility function $u(x_1, x_2) = \min\{x_1, x_2\}$.

Ex. 4 Let $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ be twice continuously differentiable. Prove that the condition $\frac{\partial}{\partial x_1} \text{MRS}_{12}(x_1, x_2) \leq 0$ is equivalent to the quasiconcavity of u as expressed by the sign criterion on the determinants of the minors of the bordered Hessian (**Theorem 1**).
